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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 06/22/2005 |
| **PI Name:** | Williams, Gordon H. M.D. | | |
| **Project Title:** | Influence of Gender and Age on Renal and Cardio-Endocrine Responses to Simulated Microgravity | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Cardiovascular Alterations Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | gwilliams@partners.org | **Fax:** | FY 617-732-5764 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 617-525-7490 |
| **Organization Name:** | Brigham and Women's Hospital | | |
| **PI Address 1:** | 221 Longwood Avenue | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Boston | **State:** | MA |
| **Zip Code:** | 02115 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 10/01/2000 | **End Date:** | 02/29/2004 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sheynberg, Natalie ( Harvard University ) | | |
| **Grant/Contract No.:** | NCC 9-58-CA00206 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Orthostatic intolerance remains an operational problem following spaceflight. In the last six years, we have addressed the subject of renal, endocrine and volume- regulating function during head-down tilt bed rest. We have found that simulated microgravity induces changes in the volume-regulating systems and electrolyte excretion, and leads to changes in cardiac electrical stability. We have also found that using our tilt-test protocol, pre-bed rest tilt-test tolerance was predictive of post-bed rest tilt-test tolerance. We propose now to conduct a double-blinded randomized trial to : 1) investigate the use of a pre-bed rest tilt-test as a means to screen countermeasures against orthostatic intolerance, and investigate the influence of these countermeasures on the renal and endocrine responses to orthostatic stress and 2) investigate the effectiveness of individualized countermeasures identified during a pre-bed rest tilt-test in reducing post-bed rest orthostatic intolerance, as well as their influence on the renal and endocrine responses to orthostatic stress. During the past several years, we have addressed the subject of renal, cardiovascular and endocrine volume-regulating factors using a simulated microgravity protocol (sixteen day-head-down tilt bed rest) in a rigorously | | |

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| | controlled environment (in-patient General Clinical Research Center with constant dietary intake, controlled activity, and regulated temperature, light exposure and sleep/wake cycles). Among the several observations that we have made on the forty individuals studied to date, the followings are key for the present proposal: Â·Initial loss of urinary volume and sodium followed by reestablishment of sodium balance through activation of the renin-angio system; Â·Continuous potassium loss; Â·A dissociation between renin and aldosterone responses to simulated microgravity and up-right tilt; Â·A decrease in autonomic function with a change towards a sympathetic-dominated system; Â·Cardiac electrical instability; Â·Subjects intolerant to head-up tilt pre-bed rest are intolerant post-bed rest, but the reverse is not true, suggesting individual susceptibility to orthostatic intolerance independent of simulated microgravity; |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The results of this study may help us better understand the pathophysiology of alterations in volume homeostatic mechanisms in cardiovascular diseases such as congestive heart failure, dysrhythmias and hypertension. |
| **Task Progress:** | During the past several years, we have addressed the subject of renal, cardiovascular and endocrine volume-regulating factors using a simulated microgravity protocol (sixteen day-head-down tilt bed rest) in a rigorously controlled environment (in-patient General Clinical Research Center with constant dietary intake, controlled activity, and regulated temperature, light exposure and sleep/wake cycles). Among the several observations that we have made on the forty individuals studied to date, the followings are key for the present proposal: Â·Initial loss of urinary volume and sodium followed by reestablishment of sodium balance through activation of the renin-angio system; Â·Continuous potassium loss; Â·A dissociation between renin and aldosterone responses to simulated microgravity and up-right tilt; Â·A decrease in autonomic function with a change towards a sympathetic-dominated system(4); Â·Cardiac electrical instability; Â·Subjects intolerant to head-up tilt pre-bed rest are intolerant post-bed rest, but the reverse is not true, suggesting individual susceptibility to orthostatic intolerance independent of simulated microgravity; Â·Finally, in a randomized double-blinded trial, we observed that midodrine was an efficient intervention for the treatment of orthostatic intolerance following simulated microgravity. |
| **Bibliography Type:** | Description: (Last Updated: 01/22/2008) |
| **Articles in Peer-reviewed Journals** | Grenon SM, Hurwitz S, Sheynberg N, Xiao X, Judson B, Ramsdell CD, Kim C, Cohen RJ, Williams GH. "Sleep restriction does not affect orthostatic tolerance in the simulated microgravity environment." J Appl Physiol. 2004 Nov;97(5):1660-6. PMID: 15234956 , Nov-2004 |
| **Articles in Peer-reviewed Journals** | Grenon SM, Hurwitz S, Xiao X, Sheynberg N, Ramsdell CD, Kim C, Cohen RJ, Williams GH. "Readaptation from simulated microgravity as a stimulus for improved orthostatic tolerance: role of the renal, cardioendocrine, and cardiovascular systems." J Investig Med. 2005 Mar;53(2):82-91. PMID: 15810494 , Mar-2005 |
| **Articles in Peer-reviewed Journals** | Grenon SM, Xiao X, Hurwitz S, Ramsdell CD, Sheynberg N, Kim C, Williams GH, Cohen RJ. "Simulated microgravity induces microvolt T wave alternans." Ann Noninvasive Electrocardiol. 2005 Jul;10(3):363-70. PMID: 16029389 , Jul-2005 |
| **Awards** | Grenon M. "Our post-doctoral fellow, Dr. Marlene Grenon, was included in a book on women in space careers: Women of Space: Cool Careers on the Final Frontier (Apogee Books Space Series), by Laura S. Woodmansee, 2004. " Jan-2004 |